

Robin Ferincz, PhD

Biophysical Engineer



CONTACT

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Nationalities:

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Swiss residence permit:

Permit B EU/EFTA

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LANGUAGES

- German - First language
- English - Advanced
- French - Intermediate

PROFILE

Engineer with five years of expertise in medical imaging, specializing in data processing, image reconstruction, AI development, and simulation-driven research, complemented by interdisciplinary experience in robotics, materials science, and applied physics.

EXPERIENCE

Research Fellow – *UNIL* - Lausanne

03.2022 – 03.2026

- Developed an automated simulation and reconstruction framework for motion-robust fetal cardiac MRI, combining image registration, compressed sensing, and quantitative validation. (MATLAB, Python, C++)
- Developed deep-learning models for automated anatomical landmark detection in fetal MRI using annotated clinical datasets and domain-randomized synthetic training data. (MATLAB, PyTorch)
- Co-authored 14 peer-reviewed publications and presented research at 10 international conferences, communicating technical results to multidisciplinary audiences.

Research Fellow – *JMU Würzburg / TU Munich* - Würzburg

08.2021 – 02.2022

- Non-invasive in-ovo gender identification of chicken embryos with MRI.
- Developed a semi-automated Python pipeline for quantitative MRI image analysis, reducing manual processing and improving reproducibility. (Python, OpenCV)
- Designed MRI phantoms and coordinated measurement schedules.

Internship Supervisor – *JMU Würzburg* - Würzburg

10.2020 – 04.2021

- Supervised hands-on experimental research in practical physics courses, instructing over 30 students on technical execution, data acquisition, and analysis in electronics, calorimetry, and NMR spectroscopy.

Medical Station Assistant – *Missionsärztliches Institut* - Würzburg

11.2019 – 03.2020

- Supported nursing staff in maintaining sterile clinical environments and assisting with patient care in a premature baby ward.

Research Internship – *Fraunhofer ISC* - Würzburg

08.2018

- Characterized electrochromic thin films using spectrometry, cyclic voltammetry, and galvanostatic cycling under inert-atmosphere conditions.
- Fabricated experimental samples and analyzed measurement data. (Python)

Working Student – *EmQopter GmbH* - Würzburg

11.2017 – 02.2019

- Supported the development, testing, and maintenance of UAV systems.
- Manufactured and assembled UAV components using CNC machining, CAD design (Autodesk Inventor), additive manufacturing, and THT/SMD electronics assembly.

Research Intern – *Wöhlk Contactlinsen GmbH* - Schönkirchen

08.2017 – 10.2017

- Supported research, development, and production activities for contact lens materials.
- Performed quality control and material characterization of contact lenses.

Intern – *EmQopter GmbH* - Würzburg

05.2017 – 07.2017

- Insight into hardware and software development for UAV systems. (C++)

Sports Retail Salesman – *Decathlon* - Würzburg

07.2015 – 03.2017

- Delivered customer service and technical product advice in a fast-paced retail environment.

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AREAS OF EXPERTISE

- Medical image reconstruction & quantitative data processing
- AI / deep learning / machine learning
- Simulation-driven research
- Motion correction (registration, artifact detection/rejection)
- Compressed sensing reconstruction
- Scientific writing & international presentation
- Interdisciplinary work: robotics, materials science, applied physics

INTERESTS

- Guitar
- Basketball
- Drones (A2 license)

EDUCATION

PhD in Biophysical Engineering

03.2022 – 03.2026

UNIL – Lausanne, Switzerland

Dissertation: Developing motion robust volumetric dynamic fetal cardiac Magnetic Resonance Imaging

- Developed algorithms for motion-robust 4D (3D+time) MRI reconstruction and quantitative image analysis.
- Combined physics-based numerical simulations with clinical MRI data for algorithm development and validation.
- Developed retrospective motion correction methods using image registration and artifact detection/rejection.
- Applied compressed sensing techniques for accelerated volumetric MRI reconstruction.
- Designed deep-learning pipelines for automated anatomical landmark detection.
- Authored peer-reviewed scientific publications and presented research at international conferences.

B. Sc. + M. Sc., Nanostructure-Technology

10.2014 – 02.2022

Julius-Maximilians-University – Würzburg, Germany

- Broad interdisciplinary education spanning physics, materials science, chemistry, semiconductor technology, optics, spintronics, and magnetism.
- Strong foundation in experimental methods, scientific computing, statistics, and uncertainty analysis.

Master thesis: Characterization of myelin-water for enhanced myelin-MRI at 3 Tesla

Bachelor thesis: Polariton condensation in hemispherical microcavities with embedded fluorescent protein

Exchange Semester

04.2020 – 10.2020

Kanazawa University – Kanazawa, Japan

International exchange program with coursework in engineering and applied physics as well as cultural and language studies.

AWARDS

- Awarded four educational travel stipends for international scientific conferences (3× ISMRM, 1× SCMR)
- Suma Cum Laude Merit Award at the ISMRM 2025
- Finalist for the 2026 JCMR Pohost Prize

PUBLICATIONS

- **Automatic Respiratory and Bulk Patient Motion Corrected 3D Fetal MRI**, Ferincz R, Alamo L, Secinaro A, Tenisch E, Bustin A, Piccini D, Prša M, Yerly J, Roy C W. *Magnetic Resonance in Medicine*. 95(4), 1972-1985. 2026 Apr. doi:10.1002/mrm.70166.
- **Automatic respiratory and bulk patient motion corrected (ACROBATIC) free-running whole-heart 5D MRI**, Ferincz R, Prša M, Tenisch E, Yerly J, Roy C W. *Journal of Cardiovascular Magnetic Resonance Imaging*. 28(1). 2026. doi:10.1016/j.jocmr.2025.102673.
- **Fully automated motion-corrected 2D CINE MRI of the fetal heart**, Ferincz R, Ogier A C, Romanin L, Alamo L, Tenisch E, Prša M, Yerly J, Seed M, Macgowan C, Stuber M, Roy C W. *CMR 2024 Scientific Sessions*, London. doi:10.1016/j.jocmr.2024.100198.
- Co-author of 14 peer-reviewed journal and conference publications in medical imaging and MRI reconstruction.
- Research presented at 10 international conferences, including ISMRM and SCMR.
- Google scholar:
<https://scholar.google.com/citations?user=XOjy8GIAAA>